

Math_3MB3_Project

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Section 1: Introduction

With the development of medical science, personalized medicine has gradually become an important strategy for the treatment of various diseases. In personalized medicine, the effect of drug therapy not only depends on the nature of the drug itself, but is also significantly affected by the mode of administration and metabolic pathway. Therefore, understanding how different modes of administration and metabolic pathways affect the actual dosage of a drug in a patient's body is important for optimizing drug treatment regimens, reducing side effects, and improving efficacy.

Drug delivery and metabolism are key factors in medicine and pharmacology. There are a variety of delivery methods that can be used to treat a patient, and the metabolic pathway will vary depending on the mode of delivery and the type of drug. Due to the individual variability of patients, the mode of administration chosen will vary. For example, oral, intravenous, subcutaneous, etc., each modality has its own unique pharmacokinetic and pharmacodynamic characteristics that affect the absorption, distribution, and metabolic processes of the drug. (Adepu & Ramakrishna, 2021).

A key challenge in the process of drug therapy is the need to go for controlling the dose of the drug in the body. This dose must be large enough to be effective in treating the patient, but not too much to avoid side effects or danger. Conventional modes of drug delivery (tablets, capsules, syrups, ointments, etc.) have low bioavailability, fluctuating plasma drug concentrations, and do not allow for sustained release. (Adepu & Ramakrishna, 2021) In this case, intravenous injection can achieve sustained release. Different modes of administration and routes of metabolism can have a significant impact on the dose of a drug in the body, so it is important to investigate how these factors can safely and effectively treat patients. This project, out of curiosity about the fundamentals of pharmacology, will focus on the effects of different modes of administration and routes of metabolism on the concentration of a drug in a patient's body. There will be a focus on analyzing two modes of drug administration, oral drugs and intravenous injections. We would like to build simple models to simulate how the metabolism of a drug in the human body would change under different modes of administration.

Our project will focus on answering the question: how do different modes of administration and metabolic pathways affect the concentration of a drug in a patient's body? In order to answer this question, we will discuss hypotheses and develop mathematical models as well as plots via R. In addition, in the next articles we will analyze and discuss the effects of different variables on drug absorption in the human body. Our aim is to improve the efficacy of the drug as well as mitigate the negative effects of the drug on the human body.

Section 2: Modelling

2.1 Base model

2.1.1 Introduction for base model

In our upcoming study, we are focusing on simulating the pharmacokinetics of intravenous injections and oral tablets to study drug absorption and metabolism in adults. We recognize that several factors influence metabolic rate, including age, weight, gender, genetic background, and liver and kidney health. Therefore, our model uses average data from a broad demographic of Canadian adults.

The primary mechanisms of drug delivery modeled in our study are intravenously and orally administered drugs. Intravenous administration allows drugs to bypass the digestive system and hepatic metabolism, thus providing immediate drug availability in the circulatory system. In contrast, oral drug delivery involves a complex absorption process that is influenced by the gastrointestinal environment and hepatic first-pass metabolism.

The rate at which the body processes drugs in the real world depends on two factors: the rate at which the drug is administered and the rate at which the drug is processed. We can imagine the human body as a container, so the amount of drug contained in the body is:

current amount of drug in the body = inflow - outflow

Let $\mathbf{A}(\mathbf{t})$ be the amount of a drug in the body (mg) at time $\mathbf{t}$ (hours). Then we can denote the dosing rate $\mathbf{D}(\mathbf{t})$ (mg/hour), which is independent of the current amount of drug in the body $\mathbf{A}(\mathbf{t})$, and the processing rate $\mathbf{P}(\mathbf{A})$ (mg/hour), which does depend on $\mathbf{A}$. Then the model for $\mathbf{A}(\mathbf{t})$ can be cast as an “inflow minus outflow” relationship:

$$\frac{dA}{dt} = D(t) - P(A)$$

We assume that there is no drug in the body at first ($\mathbf{A}(\mathbf{0})=\mathbf{0}$).

The rate of drug elimination is concentration dependent. This means that as the concentration of the drug increases, the rate of drug elimination increases accordingly and the rate of elimination is directly proportional to the concentration of the drug. So we can assume that the metabolism of the drug is linear("*Pharmacokinetics and Metabolism in Drug Design*"). We can set $\mathbf{c}$ as the clearance rate(mg/hour), so we have:

$$P(A) = cA$$

To sum up, our model simplifies these complex real-world processes by assuming:

Linear metabolism(the metabolism of the drug is linear)

Immediate drug dispersion(We assume that the drug spreads rapidly through the bloodstream)

Neglected variability(Individual variability in metabolism due to genetic differences or health status is not modeled to maintain focus on average responses)

No drug in the body before

In this study, we have selected four drugs that enter the body at the same concentration. The different metabolic rates of these drugs in the human body will be the variable in this research. The four drugs chosen for the experiment are Meloxicam, Hydrocortisone, Ibuprofen and Ketamine. According to the *National Library of Medicine*, the metabolic rates of these four drugs in the human body are 0.625 mg/hour, 1 mg/hour, 67 mg/hour, and 100 mg/hour, respectively.

2.1.2 Intravenous Plan

Below we mainly explain intravenous injection. Intravenous injection is a medical operation in which drugs or other fluids are injected directly into a vein, a method that allows for the rapid delivery of drugs throughout the body or to specific sites. This is because substances injected into a vein can enter the circulatory system directly, bypassing the digestive system and primary metabolism in the liver, which allows for rapid drug delivery.

In intravenous injection, we can assume that a total of D_{tot} mg of the drug is given intravenously at a constant rate of r mg/hour for the first h hours, at which point drug administration ceases:

$$D(t) = \begin{cases} r & \text{if } 0 \leq t \leq h \\ 0 & t > h \end{cases}$$

The time length of intravenous injection can vary depending on a number of factors, including the type of drug injected, the dose, and the desired therapeutic outcome. Based on the statistics from the "*Journal of Pharmacology and Experimental Therapeutics*" regarding recommended injection times and injection rate for commonly used intravenous drugs, we will assume $h = 1$ hour and $r = 100$ mg/hour in our study.

If we plug $D(t)$ and $P(A)$ in the first differential equation, we can get drug processing rate of the body is:

$$\frac{dA}{dt} = \begin{cases} r - cA & \text{if } 0 \leq t \leq h \\ -cA & t > h \end{cases}$$

After solving the linear first-order ordinary differential equation and the autonomous ordinary differential equation. We can get the function of the amount of a drug in the body is:

$$A(t) = \begin{cases} \frac{r}{c} - (\frac{r}{c} - A(0))e^{-ct} & \text{if } 0 \leq t \leq h \\ A(h)e^{-ct} & t > h \end{cases}$$

Plugging in the $A(0)$, r and h value we have assumed. we get:

$$A(t) = \begin{cases} \frac{100}{c} - (\frac{100}{c} - 0)e^{-ct} & \text{if } 0 \leq t \leq 1 \\ A(1)e^{-ct} & t > 1 \end{cases} \quad (1)$$

2.1.3 Dosage Plan

Oral drug administration is one of the most common ways of drug delivery as the drug is introduced into the body through the digestive tract. Oral drug delivery is widely used because of its convenience, low cost, and good patient compliance. Assume the drug is given as a pill that is designed to dissolve slowly. As the pill dissolves, less and less of the medication is released. So we can set dosing rate be:

$$D(t) = D_{max}e^{-\frac{t}{h}}$$

So, we can get drug processing rate of the body is:

$$\frac{dA}{dt} = D_{max}e^{-\frac{t}{h}} - cA$$

When $t = 0$,

$$D(0) = D_{max}e^{-\frac{0}{h}}$$

$$D(0) = D_{max} \times 1$$

$$D(0) = D_{max}$$

we can get :

$$D(t) = D(0)e^{-\frac{t}{h}}$$

Then, drug processing rate of the body will be:

$$\frac{dA}{dt} = D(0)e^{-\frac{t}{h}} - cA$$

Solving it after integrating, we get the function of the amount of a drug in the body is:

$$A(t) = \frac{D(0)}{c - \frac{1}{h}}e^{-\frac{t}{h}} + \left(A(0) - \frac{D(0)}{c - \frac{1}{h}}\right)e^{-ct}$$

We will also assume $D(0)$ to be 100mg, $A(0)$ is 0mg and $h = 1$ hour. So we have:

$$A(t) = \frac{100}{c - 1}e^{-t} + \left(0 - \frac{100}{c - 1}\right)e^{-ct} \quad (2)$$

2.2 Analysis of Base Model

Stating from our model (1). To find the equilibrium point, we need to set $\frac{dA}{dt} = 0$.

So we have:

$$r - c\dot{A} = 0$$

$$\dot{A} = r/c$$

So our equilibrium is $\frac{r}{c}$

Check for stable:

$$\frac{d}{dA} \frac{dA}{dt} = -c$$

Since $c > 0$, $-c < 0$. We have the result that the equilibrium is stable.

After that we need to set up the plot for the four drugs chosen which are Meloxicam, Hydrocortisone, Ibuprofen and Ketamine. The metabolic rates of these four drugs in the human body are 0.625 mg/hour, 1 mg/hour, 67 mg/hour, and 100 mg/hour, respectively. Plugging in this c value in model (1). Also in R coding, we can use Euler's method to check our ODE solving.

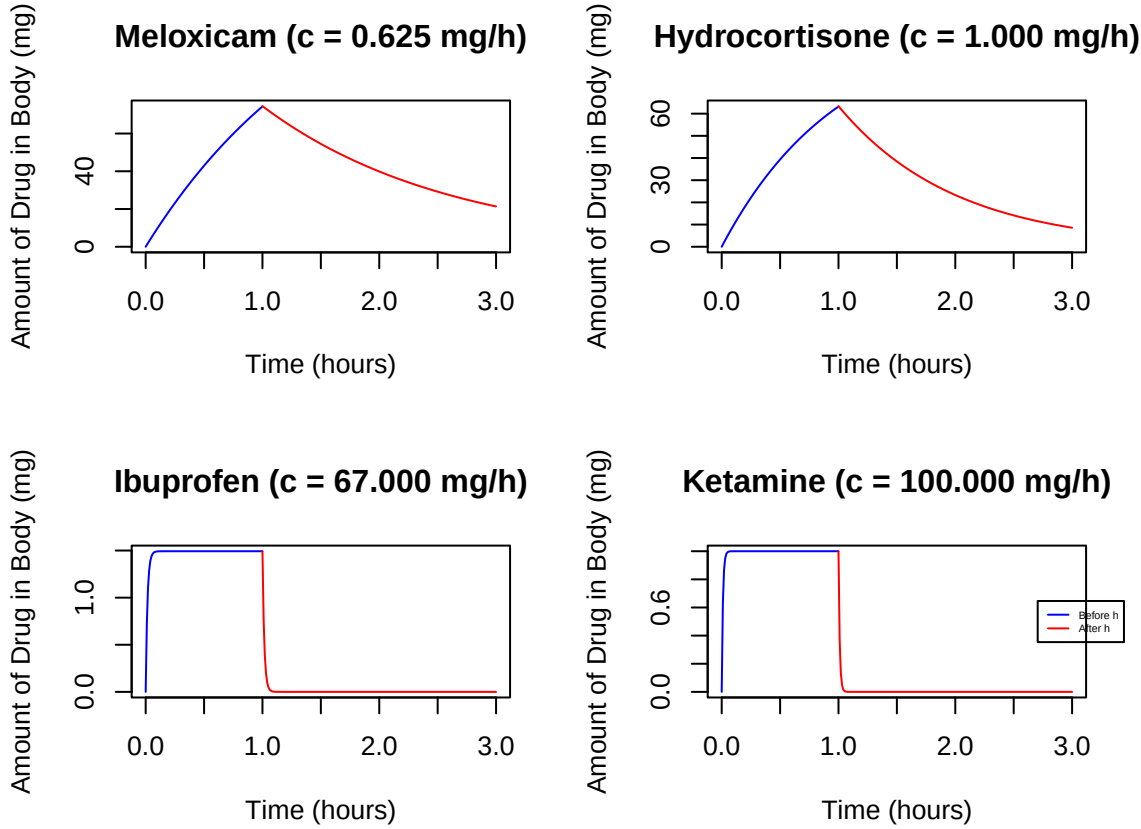


Figure 1: Figure 1: Intravenous injection basic model

This plot shows us the concentrations of the four drugs within three hours of entering the body by the intravenous route. The junction of the red and blue lines in each graph indicates the h time, the point at which drug administration ceased. At this time point, there is a significant change in the concentration of each drug, jumping directly from the rate of administration to the elimination process.

For Meloxicam and Hydrocortisone, the concentration of the drug increases gradually during administration (blue line), indicating that the rate of accumulation of the drug concentration is greater than the rate of its elimination during administration. Therefore, their concentration rises with the duration of administration.

For Ibuprofen and Ketamine, the concentration of the drug remains constant during administration (blue line), indicating that a stable concentration is maintained during administration. And there is some difference between the two before. For example, the concentration of Ibuprofen ended up staying around 1.4mg, while the concentration of Ketamine stayed at 1mg.

From this we can give a guess that the concentration of the drug at the peak of its stay in the body can be changed by varying the rate of injection.

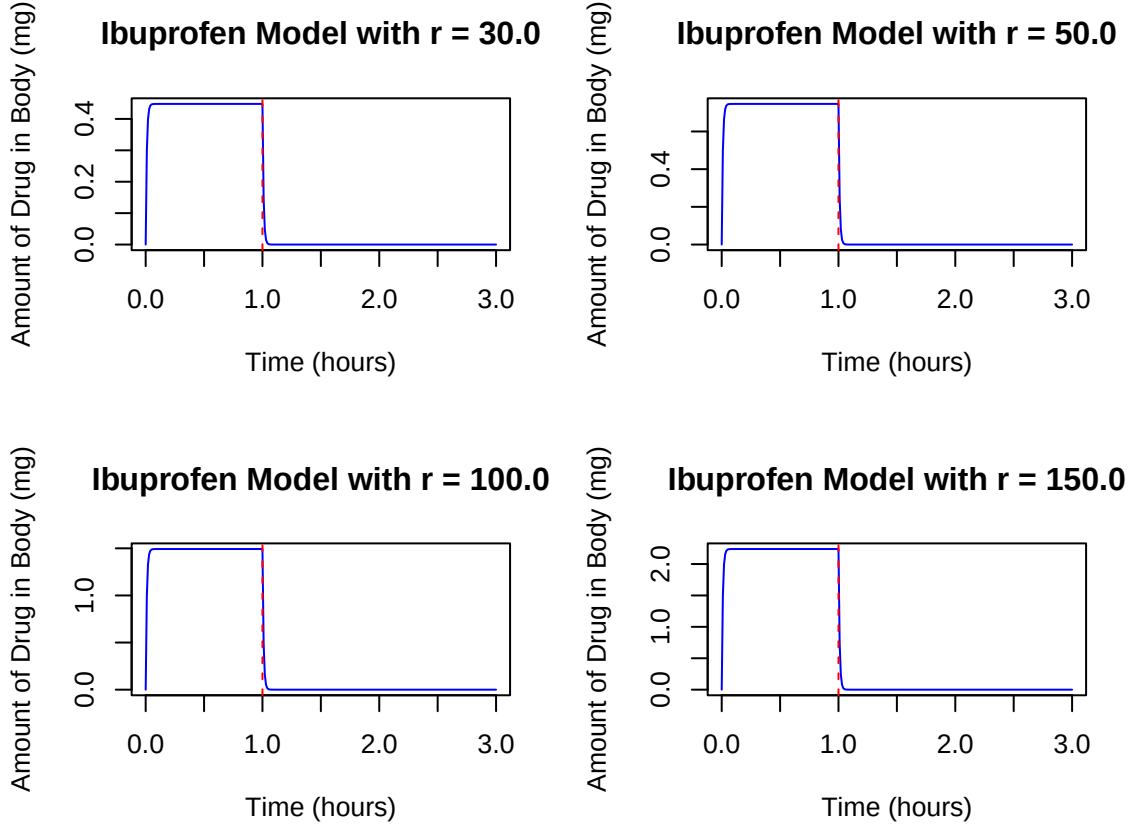


Figure 2: Figure 2: focus on Ibuprofen

From this plot, we can observe that changing the intravenous injection rate of ibuprofen alters the peak residual concentration of the drug in the body. Thus, adjusting the injection speed to control the drug's residual concentration is a viable strategy. This is very useful in clinical settings for designing drug administration plans. *The National Library of Medicine* indicates that the effective concentration of ibuprofen for patients with low fever is 1.7 mg. Therefore, we can control the injection rate at 150 mg/hour to maintain the drug at an effective concentration in the body over a prolonged period.

For our model (2). To find the equilibrium point, we need to set $frac{dA}{dt} = 0$.

$$0 = D_{max}e^{-\frac{t}{h}} - cA$$

$$\dot{A} = \frac{D(0)e^{-t/h}}{c}$$

So our equilibrium is $\dot{A}$

Check for stable:

$$\frac{d}{dA} \frac{dA}{dt} = -c$$

Since $c > 0$, $-c < 0$. We have the result that the equilibrium is stable.

We will also use R to plot the data with different drugs and different c . Since we have assumed that $D(0) = 100\text{mg}$, and we have solve the ODE in previous section. So we can just use the $A(t)$ formula for dosage plan to plot the data.

In this model, it is necessary for the clearance rate c to be unequal to $1/h$. If they are equal, it would lead to a division by zero in the $A(t)$ equation, which would result in the absence of a real number solution. In analyzing Hydrocortisone, c is 1 mg/h and h equals 1 hour . To address this issue, we have adjusted c to 1.0001 mg/h . This adjustment avoids the problem of no solution for $A(t)$, although it introduces a minor error.

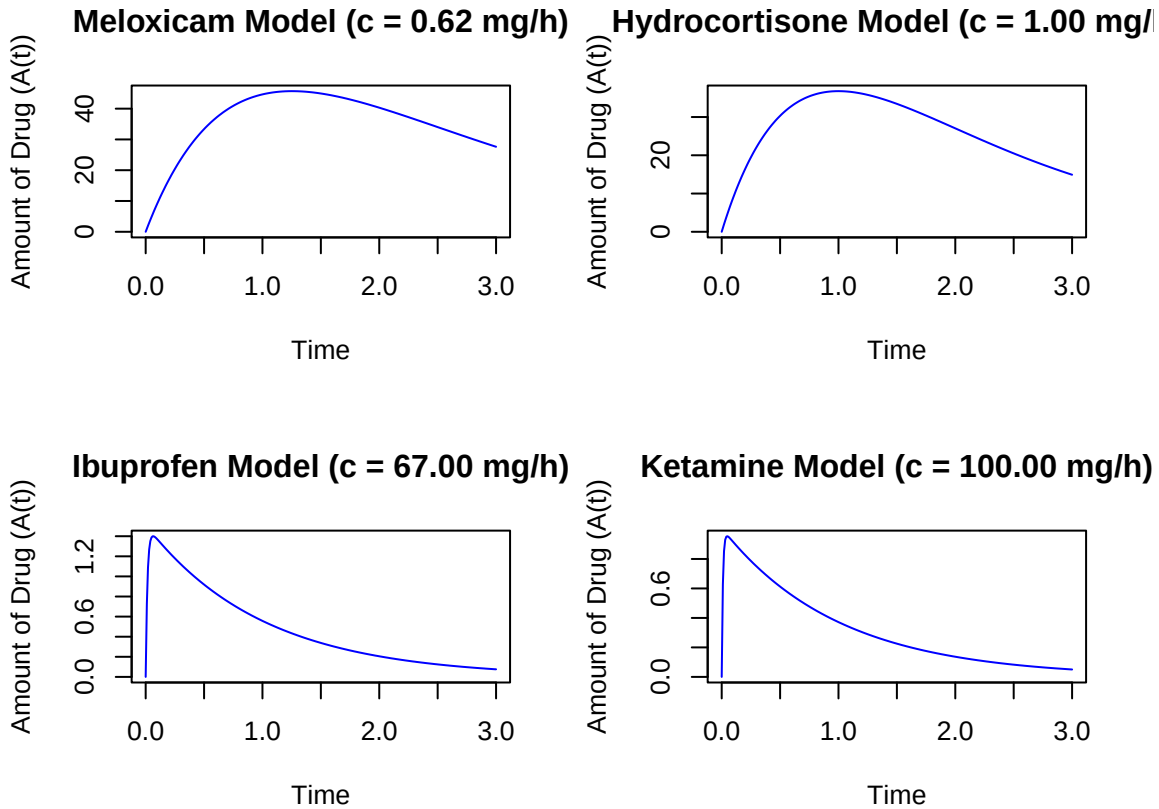


Figure 3: Figure 3: drug is given as a pill

By comparing plot1 and plot3 we can conclude the following:

Intravenous administration shows a rapid rise in drug concentration to a peak and then a gradual decline, reflecting the fact that intravenous administration can rapidly increase the concentration of drug in the blood.

The graphs for oral administration show a gradual increase in drug concentration, which may indicate that the absorption process is slower and not as rapid as intravenous administration.

Peak concentrations were generally higher for intravenous administration than for oral administration.

There is no clear peak in the model for oral administration, and drug concentrations rise or fall smoothly.

So in reality for situations that require a rapid response, such as pain relief or anti-inflammation, intravenous administration may be more appropriate. For treatments requiring long-term maintenance of effective concentrations, oral administration may be preferable because it provides a more gradual release of the drug.

2.3 Model Extension

After in-depth study and analysis of the dynamic distribution of drugs in the human body, we have performed a series of optimizations to the model to make it more closely match the dynamic behavior of drugs in the real world. Numerous studies have shown that the binding of drugs to proteins in plasma and their distribution in blood cells and intercellular matrix are closely correlated with drug concentration, and this correlation is particularly significant when drug concentration changes. In practice, the total concentration of a drug in blood or serum does not fully reflect the pharmacokinetic properties of a drug because these properties may change with varying concentrations. Metabolic rate-concentration interdependence occurs when the drug concentration reaches a high enough value to saturate enzymes or transporter proteins involved in the metabolic process. Taking certain drugs, the way they bind and distribute in the body varies with concentration, which affects their metabolic rate. Warfarin, for example, as a commonly used anticoagulant, has a pharmacokinetic behavior that varies with concentration, which in turn has an impact on its efficacy and safety. Similarly, the in vivo metabolism of drugs with a narrower range of efficacy, such as phenytoin, also exhibits saturation kinetics, where small changes in dosage may cause large fluctuations in the concentration of the drug in vivo, which requires the clinician to be extra cautious in its use, to closely monitor the patient's level of the drug and adjust the dosage in a timely manner.

In this enhanced model, we retain the core definitions from the initial framework: $P(A)$, the metabolic rate, measured in mg per hour; $A(t)$, the total drug quantity within the organism, quantified in milligrams; and the clearance rate, c , in mg/h. To encapsulate the maximum capacity of the metabolic enzymes, we introduce the variable K , signifying the saturation point in milligrams. As the drug's presence nears this saturation point, the rate of metabolism gradually diminishes, ultimately approaching a standstill. In the event that the drug is undetectable within the system, the metabolic rate peaks, reaching a maximum throughput of cA .

So our model will be:

$$P(A) = cA(1 - \frac{A}{K})$$

When $t > h$,

$$\begin{aligned} D(t) &= 0 \\ \frac{dA}{dt} &= 0 - cA(1 - \frac{A}{K}) \\ f(P) &= cA(1 - \frac{A}{K}) \\ g(t) &= -1 \end{aligned}$$

solve it, we have:

$$A(t) = \frac{K}{1 + A(h)e^{ct}}$$

Also we can solve it for $0 < t < h$:

$$\frac{dA}{dt} = r - cA(1 - \frac{A}{K})$$

In the next section, we will simulate and analyze the changes brought about by the introduction of logical drug metabolism.

Section 3: Result

In logistic drug metabolism, the parameter K represents the carrying capacity of the drug metabolism system. This model extension is more realistic because the body's metabolic system does not process drugs indefinitely. Next, I will substitute different parameters and observe the behavior of the extended model and make a mathematical analysis of its equilibrium point and the stability of the equilibrium point.

$$\text{Logistic drug metabolism}(1) : \frac{dA}{dt} = \begin{cases} r - cA(1 - \frac{A}{K}), & 0 \leq t \leq h \\ -cA(1 - \frac{A}{K}), & t > h \end{cases}$$

When analyzing an extended model of intravenous injection, we first need to find the equilibrium point, where the concentration of the drug in the body no longer changes $\frac{dA}{dt} = 0$. Finding the equilibrium point can help us understand how a drug is distributed in the body under specific drug administration rates and drug processing rates.

We first find the equilibrium points when $0 \leq t \leq h$. By solving the following equation: $\frac{dA}{dt} = r - cA(1 - \frac{A}{K}) = 0$. We found two equilibrium points.

$$A^* = \begin{cases} \frac{1}{2}k + \sqrt{-\frac{r}{c}k + \frac{1}{4}k^2} \\ \frac{1}{2}k - \sqrt{-\frac{r}{c}k + \frac{1}{4}k^2} \end{cases}$$

From the lecture, we learned how to determine the stability of logistic model. To check stability, we first need to compute $\frac{d}{dA}(\frac{dA}{dt})$.

$$\frac{d}{dA}(\frac{dA}{dt}) = -c + \frac{2c}{k}A$$

When $A^* = \frac{1}{2}k + \sqrt{-\frac{r}{c}k + \frac{1}{4}k^2}$, $f'(A^*) = \sqrt{-\frac{r}{c}k + \frac{1}{4}k^2}$, which is unstable. When the drug concentration reaches this equilibrium point in the body during administration, changes near the equilibrium point will move the system away from equilibrium, so this equilibrium point is unstable.

When $A^* = \frac{1}{2}k - \sqrt{-\frac{r}{c}k + \frac{1}{4}k^2}$, $f'(A^*) = -\sqrt{-\frac{r}{c}k + \frac{1}{4}k^2}$, which is stable. Small changes in the system near this equilibrium point will move the system toward the equilibrium state, so this equilibrium point is stable.

Then, we need find the equilibrium points when $t > h$. By solving the following equation: $\frac{dA}{dt} = -cA(1 - \frac{A}{K}) = 0$. We found that the equilibrium points are: $A^* = 0$ and $A^* = K$. Similarly, for each equilibrium point A^* , we compute $\frac{d}{dA}(\frac{dA}{dt}) = \frac{2Ac}{K} - c$ to check the stability of the model at each point.

For $A^* = 0$:

$$f'(0) = 0 - c = -c$$

Since $c > 0$, $-c < 0$. We have the result that the equilibrium point $A^* = 0$ is stable.

For $A^* = K$:

$$f'(K) = 2c - c = c$$

Since $c > 0$. We have the result that the equilibrium point $A^* = K$ is unstable.

For $t > h$, the stable equilibrium point $A^* = 0$ represents the state with no drug in the body, while the unstable equilibrium point $A^* = K$ represents the drug concentration reaching the carrying capacity of the drug metabolism system. This shows that after the intravenous injection is stopped, the drug concentration in the body will tend to gradually decrease and eventually disappear, rather than continuing to maintain the maximum concentration.

By studying equilibrium point stability, the steady-state concentration level achieved by a drug after administration can be determined. The stability of the equilibrium point can help predict how drug content in the body will behave under different conditions. A stable equilibrium point means that the drug concentration in the body will remain stable within a certain range, while an unstable equilibrium point may lead to unpredictable behavior of the system, such as oscillations or periodic fluctuations. This is essential for proper control of drug dosage and dosing regimen.

Next, we will analyze how the amount of drug in the blood changes over time after introducing the K parameter compared with the basic intravenous injection model. Taking ibuprofen as an example, we will study how different K values and different injection rates (r) change the amount of drug in the blood after introducing logical drug metabolism.

First we set the carrying capacity (K) to 9mg and observe the changes in drug content in the body over time.

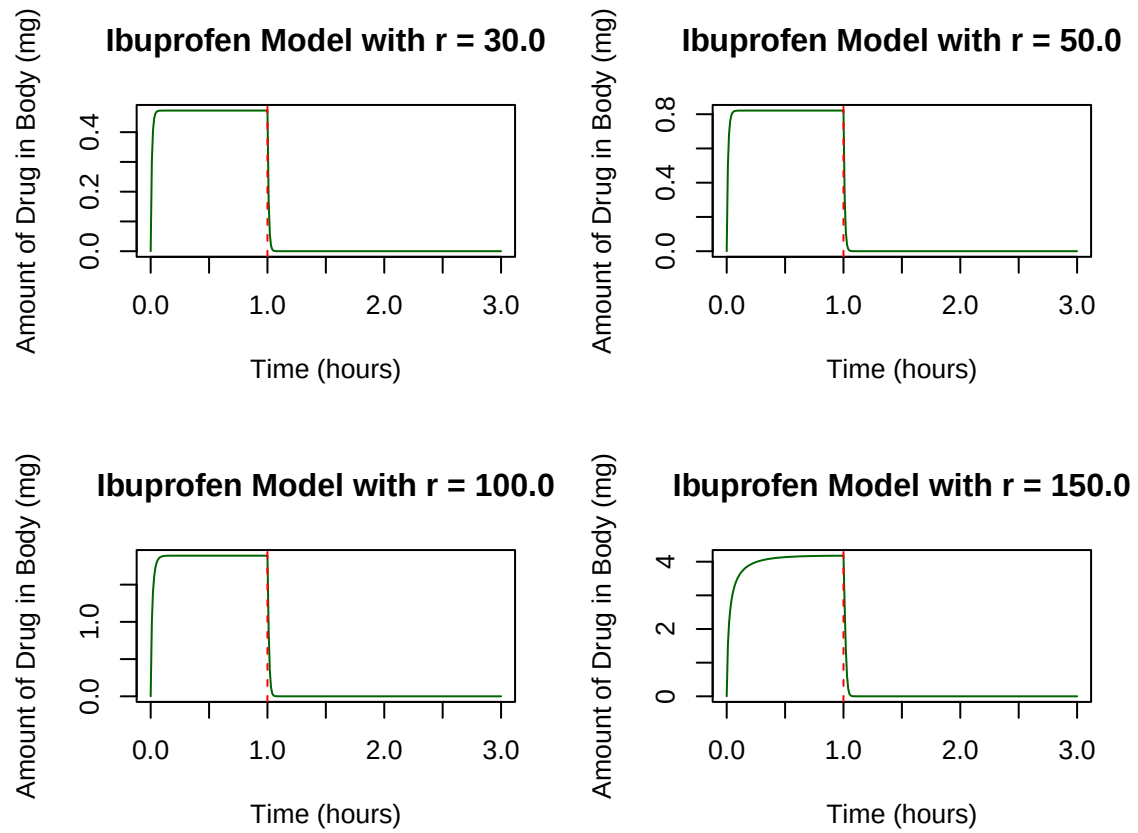


Figure 4: Figure 4: Logical Drug Metabolism (K=9)

We can find from the figure above that after the introduction of logical drug metabolism, despite the different injection rates, the maximum value of the drug concentration in the body has increased relative to the basic model. This feature is more pronounced at higher injection rates.

The reason for this result is that when the drug concentration approaches K , the drug metabolism rate will slow down, resulting in an increase in the maximum concentration of the drug in the body. In addition, at higher injection rates, the concentration of the drug increases rapidly, causing the drug concentration in the body to be closer to the value of carrying capacity (K). The saturation effect in the logistic metabolic model is better able to capture this phenomenon. Therefore, a higher injection rate will result in higher drug concentration values in the body.

We next double the K value and observe the change.

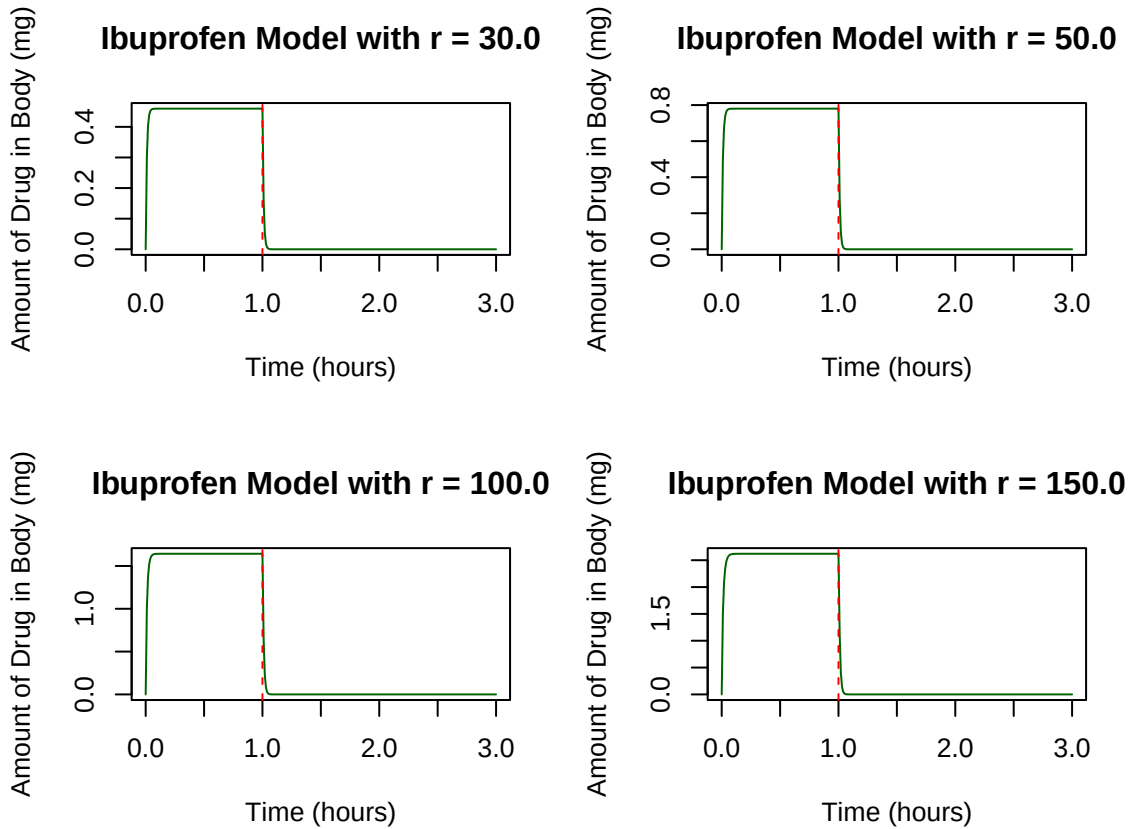


Figure 5: Figure 5: Logical Drug Metabolism ($K=18$)

From the figure above, we find that when the K value is 18 mg, the maximum drug content has also increased relative to the basic model, but it is not as significant as the lower K value. The reason for this result is that for the injection rate we selected, the K value is much higher than the drug content in the body $A(t)$. This results in the characteristics of logical metabolism not being well displayed, and the logical metabolic rate is not much different

from the metabolic rate before the K value is introduced.

We found that after introducing the concept of logical metabolism and adding the parameter K , we can see from the image that the general trend of drug content in the body has not changed significantly. However, after introducing the parameters of logical metabolism, the maximum concentration of the drug in the body generally increased under different injection rates.

Section 4: Discussion

To recap, our research question was to compare the effects of two drug delivery methods (IV and oral) and human metabolism on drug content in the body over time. This article first studies the basic models of intravenous administration and oral drugs, that is, the models without the introduction of logical metabolism. Through mathematical modeling and analysis, we can see that oral medications and intravenous methods each have advantages. The intravenous method is more effective as we have found that regardless of changes in the injection rate, the drug is able to maintain higher concentrations for a longer period of time, making it suitable for acute situations and those requiring immediate effect. As we mentioned in the modeling process, delivering ibuprofen through intravenous injection can make the drug reach the peak within an extreme time and continue at the peak to achieve analgesic effect. After the intravenous injection was stopped, what we found was that the drug concentration in the body would rapidly drop to 0 mg, thus avoiding the side effects caused by some drugs staying in the body for a long time. In addition, intravenous injection is also suitable for patients who cannot take it orally or who have absorption problems, such as patients who are unconscious or passed out. However, not all drugs desire this behavior. Through our modeling and analysis, we found that for oral drugs, the drug concentration in the body rises and decreases much slower than intravenous injection. The reason for this is that the drug is gradually absorbed in the gastrointestinal tract. Therefore, oral medications are suitable for long-term treatment, such as anti-hypertensive drugs. Additionally, oral medications require less medical assistance to administer, produce less medical waste, and are more readily available to patients.

Analysis and predictions made only through basic models are very unconvincing, because the drug processing rate of the human body in real life is not simply the multiplication of drug concentration and clearance rate. Therefore, we introduced logical metabolism as a model extension. Logical metabolism is usually more consistent with the way actual biological metabolism systems work, and therefore is more able to accurately describe the metabolic process of drugs in the body. After making changes to the basic model by introducing the K value, we found that the body's drug processing rate decreases as the drug concentration approaches the K value. Therefore, in real life, intravenously administered drugs will reach higher concentrations in the body than that in the basic model due to logical metabolism. Understanding how logical metabolism affects the maximum concentration of a drug can help doctors better adjust drug doses in real life. If administered intravenously without understanding logical metabolism, peak concentrations of the drug may increase the risk of side effects of the drug.

Moreover, it is important to note that the model used has some non-negligible limitations that may affect the interpretation and application of the study results. First, the model employs the assumption of linear metabolism, which simplifies the dynamics of the drug in vivo. While this assumption helps in the initial understanding of pharmacokinetics, it fails to capture the complex nonlinear behavior that may occur in vivo when drug concentrations change significantly. Second, the model does not incorporate metabolic rate variability due to individual genetic differences, health status, and differences in age and weight, which may limit its generalizability and accuracy across populations. In addition, the model assumes that there is no drug in the body at the start of treatment, ignoring possible sequential drug use or multi-drug interactions. This simplification, while facilitating model construction and calculation, may affect its validity and accuracy in practical clinical applications. These limitations suggest the need for caution when using the models in this study to guide clinical practice or make treatment decisions. In particular, when designing a personalized drug regimen, the model's predictions and the patient's specific situation should be considered. Such prudence not only helps to optimize treatment outcomes, but is also a sign of responsibility for patient safety. We expect that future studies will enhance the predictive ability and applicability of the model by introducing finer biokinetic parameters and considering more individual variation factors.

In summary, the model used in this project to analyze the effect of dosing method on drug concentration in the body is very basic. Therefore, as mentioned in the previous paragraph, this model has strong limitations. During the modeling process, we only considered the metabolic differences caused by different drugs. Future research should be expanded to analyze the impact of factors such as different age groups or weights on drug metabolism and absorption to further improve the model. These studies will make delivery methods more practical and take into account the diversity of drug properties. More detailed studies of the model will help determine optimal dosing strategies to improve efficacy and thus better serve clinical practice.

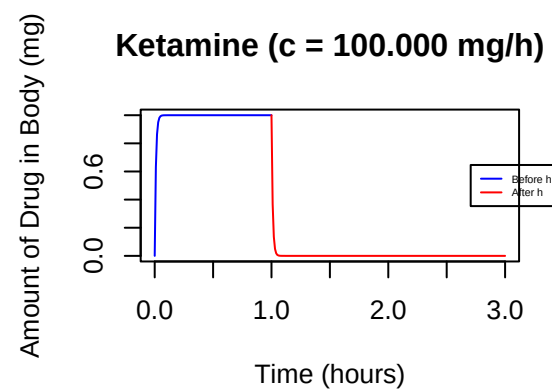
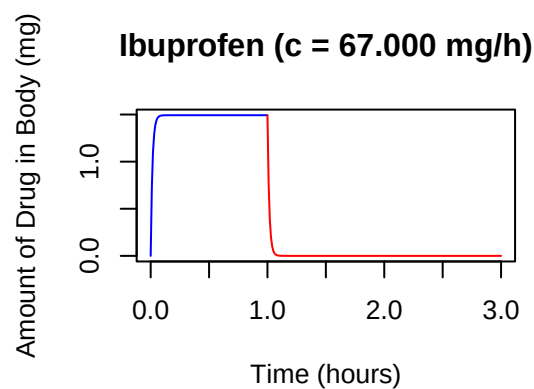
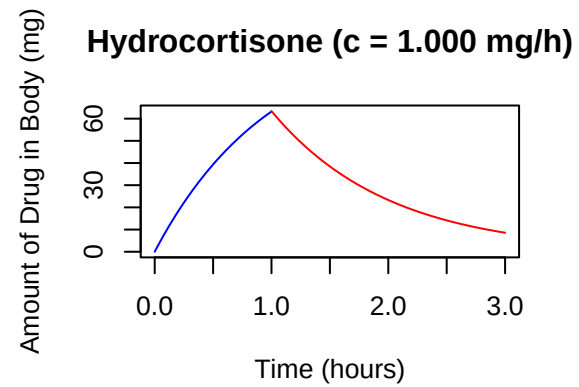
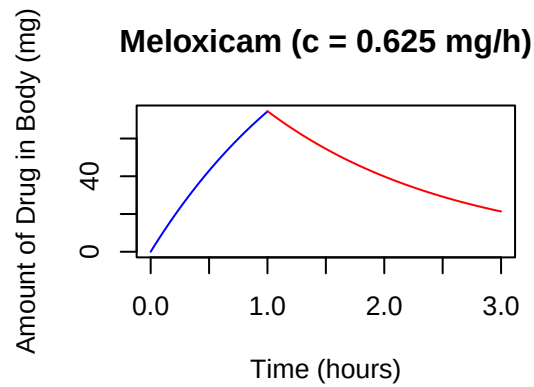
Section 5: Appendix: R Code

```
# Intravenous injection
exp_model <- function(A, r, c) {
  val <- r - c * A
  return(val)
}
exp2_model <- function(A, c) {
  val <- -c * A
  return(val)
}
h <- 1
t_vals <- seq(0, h, by = 0.01)
r <- 100
c_values <- c(0.625, 1, 67, 100)
```

```

drug_names <- c("Meloxicam", "Hydrocortisone", "Ibuprofen", "Ketamine")
par(mfrow = c(2, 2))
for (i in seq_along(c_values)) {
  c <- c_values[i]
  drug_name <- drug_names[i]
  A_vals <- rep(NA, length(t_vals))
  A_vals[1] <- 0
  for (n in 2:length(t_vals)) {
    v <- t_vals[2] - t_vals[1]
    A_vals[n] <- A_vals[n - 1] + v * exp_model(A = A_vals[n -
      1], r = r, c = c)
  }
  soln <- r/c - (r/c - A_vals[1]) * exp(-c * t_vals)
  t_val <- seq(h, 3, by = 0.01)
  A_val <- rep(NA, length(t_val))
  Con <- tail(A_vals, n = 1)
  for (n in 2:length(t_val)) {
    v <- t_val[2] - t_val[1]
    A_val[n] <- A_val[n - 1] + v * exp2_model(A = A_val[n -
      1], c = c)
  }
  soln2 <- Con * exp(-c * (t_val - h))
  plot(t_vals, soln, type = "l", col = "blue", xlab = "Time (hours)",
    ylab = "Amount of Drug in Body (mg)", main = sprintf("%s (c = %.3f mg/h)",
      drug_name, c), xlim = range(c(t_vals, t_val)), ylim = range(c(soln,
      soln2)))
  lines(t_val, soln2, col = "red")
}
par(xpd = TRUE)
legend("right", inset = c(-0.1, 0), legend = c("Before h", "After h"),
  col = c("blue", "red"), lty = 1, cex = 0.43)

```

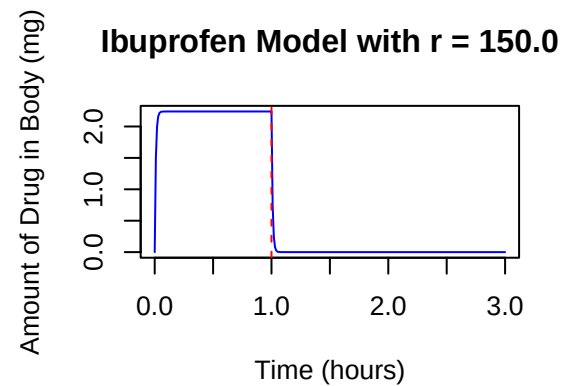
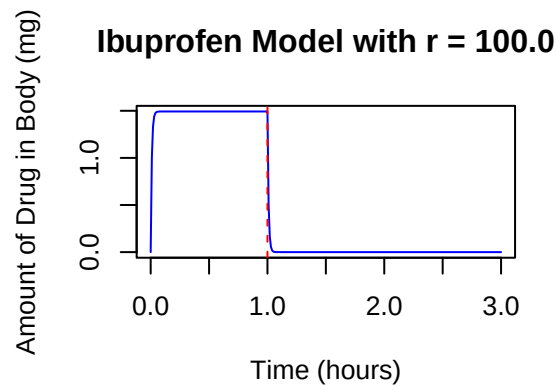
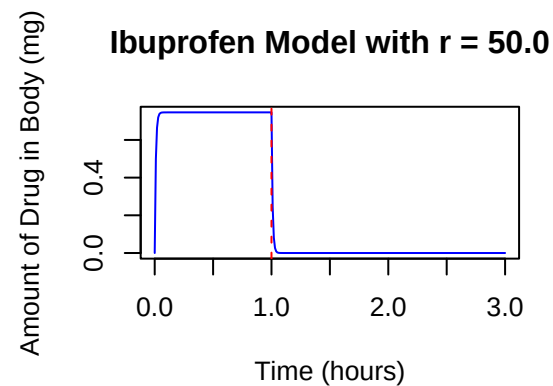
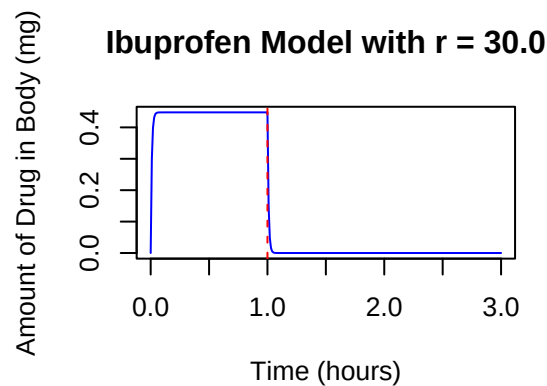



```
# Ibuprofen
exp_model <- function(A, r, c) {
  val <- r - c * A
  return(val)
}
exp2_model <- function(A, c) {
  val <- -c * A
  return(val)
}
h <- 1
t_vals <- seq(0, h, by = 0.01)
c <- 67
r_values <- c(30, 50, 100, 150)
par(mfrow = c(2, 2))
for (r in r_values) {
  A_vals <- rep(NA, times = length(t_vals))
  A_vals[1] <- 0
  for (n in 2:length(t_vals)) {
    v <- t_vals[2] - t_vals[1]
    A_vals[n] <- A_vals[n - 1] + v * exp_model(A = A_vals[n - 1], r = r, c = c)
  }
}
```

```

}
soln <- r/c - (r/c - A_vals[1]) * exp(-c * t_vals)
t_val <- seq(h, 3, by = 0.01)
A_val <- rep(NA, times = length(t_val))
A_val[1] <- A_vals[length(A_vals)]
for (n in 2:length(t_val)) {
  v <- t_val[2] - t_val[1]
  A_val[n] <- A_val[n - 1] + v * exp2_model(A = A_val[n -
    1], c = c)
}
soln2 <- A_val[1] * exp(-c * (t_val - h))
x <- c(t_vals, t_val)
y <- c(A_vals, A_val)
plot(x, y, type = "l", col = "blue", xlab = "Time (hours)",
  ylab = "Amount of Drug in Body (mg)", main = sprintf("Ibuprofen Model with r = %.1f",
    r), xlim = range(x), ylim = range(y))
abline(v = h, col = "red", lty = 2)
}

```

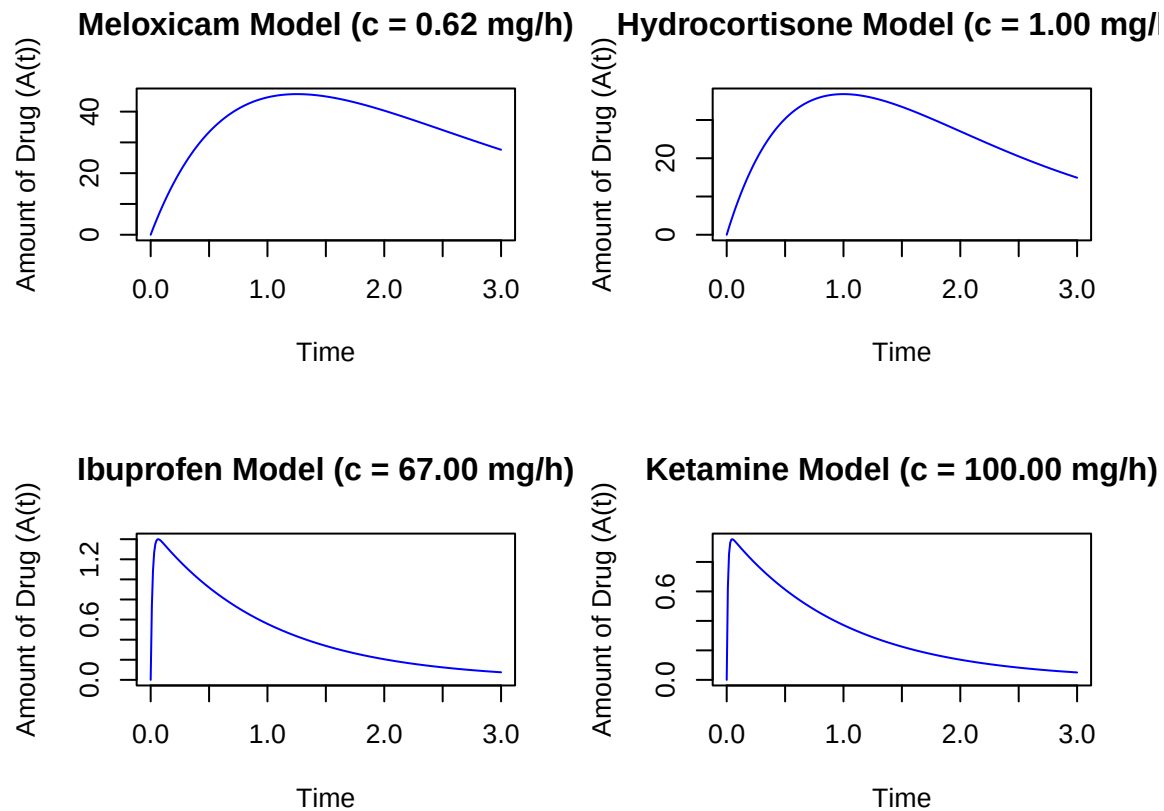


```
par(mfrow = c(1, 1))
```

```

# Oral drugs
D0 <- 100
h <- 1
dt <- 0.01
times <- seq(0, 3, by = dt)
A0 <- 0
c_values <- c(0.625, 1.001, 67, 100)
drug_names <- c("Meloxicam", "Hydrocortisone", "Ibuprofen", "Ketamine")
par(mfrow = c(2, 2))
for (idx in 1:length(c_values)) {
  c <- c_values[idx]
  drug_name <- drug_names[idx]
  A_t <- numeric(length(times))
  for (i in 1:length(times)) {
    t <- times[i]
    if (c - 1/h != 0) {
      A_t[i] <- (D0/(c - 1/h)) * exp(-t/h) + (A0 - (D0/(c -
        1/h))) * exp(-c * t)
    } else {
      A_t[i] <- NA
    }
  }
  valid_A_t <- A_t[!is.na(A_t)]
  valid_times <- times[!is.na(A_t)]
  if (length(valid_A_t) > 0) {
    plot(valid_times, valid_A_t, type = "l", col = "blue",
      xlab = "Time", ylab = "Amount of Drug (A(t))", main = sprintf("%s Model (c = %.2f
        drug_name, c), ylim = range(valid_A_t, na.rm = TRUE))
  } else {
    plot(0, 0, type = "l", col = "blue", xlab = "Time", ylab = "Amount of Drug (A(t))",
      main = sprintf("%s Model (c = %.2f mg/h) - Not computable",
        drug_name, c))
  }
}

```



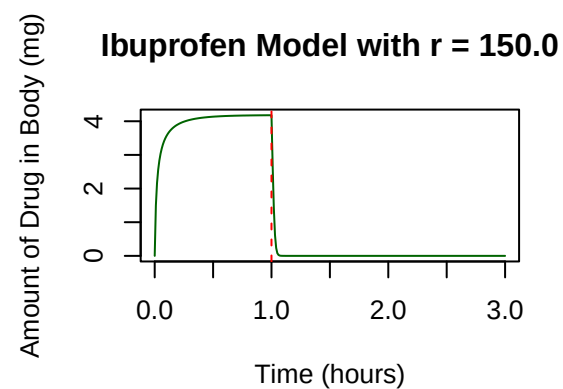
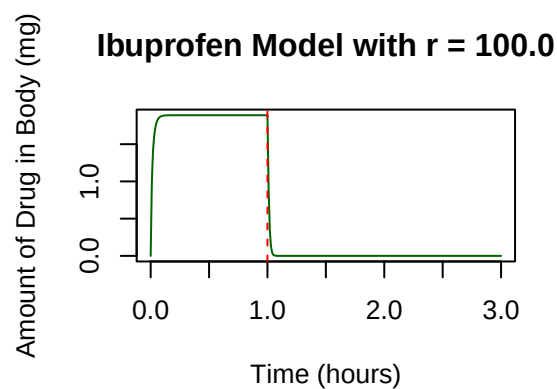
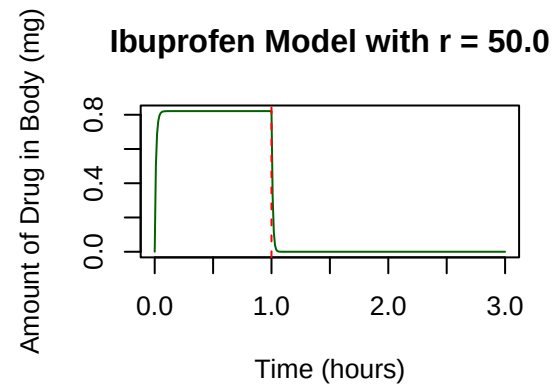
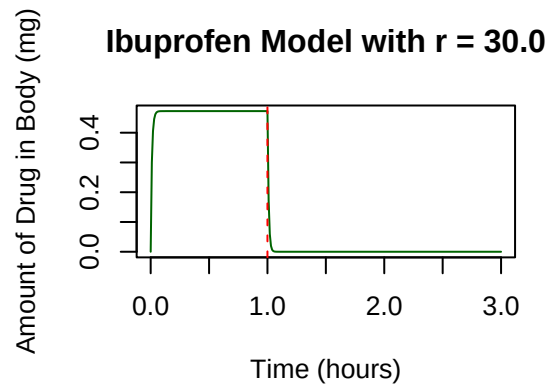
```
par(mfrow = c(1, 1))

# Logical Drug Metabolism (1)
exp_model <- function(A, r, c, K) {
  val <- r - c * A * (1 - A/K)
  return(val)
}
exp2_model <- function(A, c, K) {
  val_1 <- -c * A * (1 - A/K)
  return(val_1)
}
h <- 1
K <- 9
t_vals <- seq(0, h, by = 0.01)
c <- 67
r_values <- c(30, 50, 100, 150)
par(mfrow = c(2, 2))
for (r in r_values) {
  A_vals <- rep(NA, times = length(t_vals))
  A_vals[1] <- 0
  for (n in 2:length(t_vals)) {
```

```

    v <- t_vals[2] - t_vals[1]
    A_vals[n] <- A_vals[n - 1] + v * exp_model(A = A_vals[n -
        1], K = K, r = r, c = c)
}
soln <- r/c - (r/c - A_vals[1]) * exp(-c * t_vals)
t_val <- seq(h, 3, by = 0.01)
A_val <- rep(NA, times = length(t_val))
A_val[1] <- A_vals[length(A_vals)]
for (n in 2:length(t_val)) {
    v <- t_val[2] - t_val[1]
    A_val[n] <- A_val[n - 1] + v * exp2_model(A = A_val[n -
        1], c = c, K = K)
}
soln2 <- A_val[1] * exp(-c * (t_val - h))
x <- c(t_vals, t_val)
y <- c(A_vals, A_val)
plot(x, y, type = "l", col = "darkgreen", xlab = "Time (hours)",
    ylab = "Amount of Drug in Body (mg)", main = sprintf("Ibuprofen Model with r = %.1f",
        r), xlim = range(x))
abline(v = h, col = "red", lty = 2)
}

```



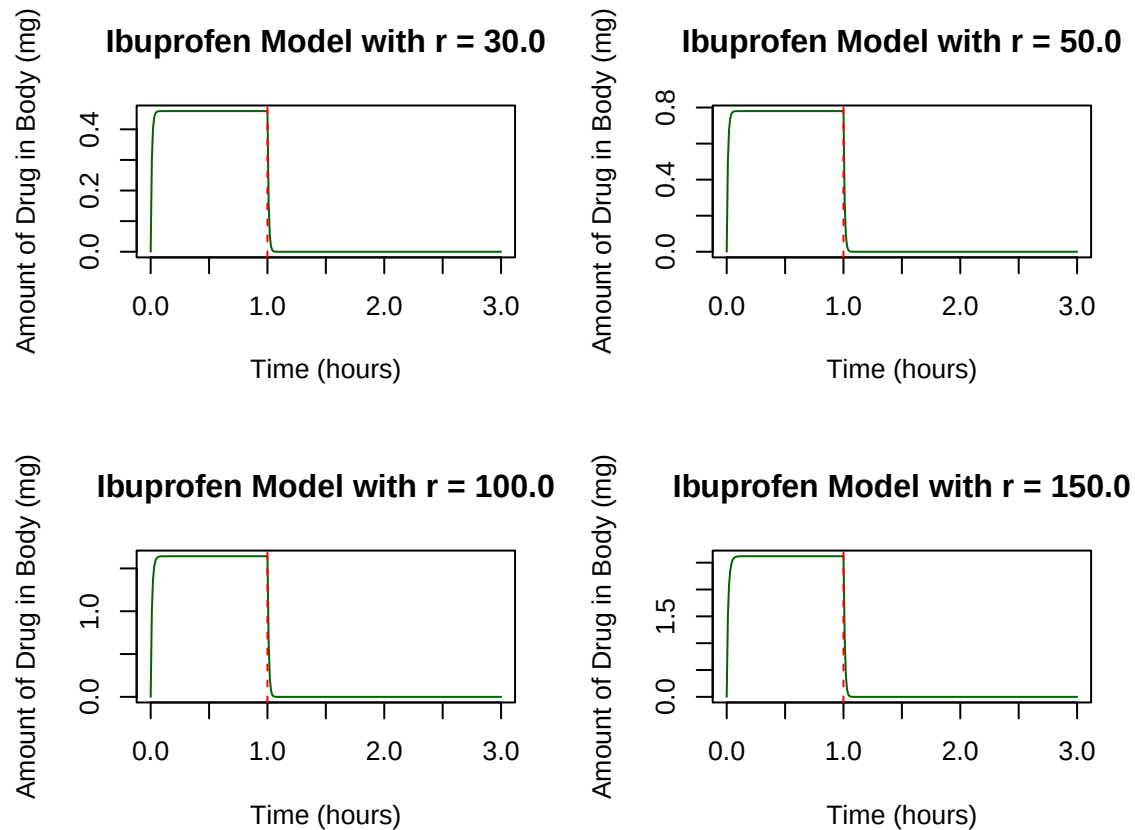
```
par(mfrow = c(1, 1))

# Logical drug metabolism (2)
exp_model <- function(A, r, c, K) {
  val <- r - c * A * (1 - A/K)
  return(val)
}
exp2_model <- function(A, c, K) {
  val_1 <- -c * A * (1 - A/K)
  return(val_1)
}
h <- 1
K <- 18
t_vals <- seq(0, h, by = 0.01)
c <- 67
r_values <- c(30, 50, 100, 150)
par(mfrow = c(2, 2))
for (r in r_values) {
  A_vals <- rep(NA, times = length(t_vals))
  A_vals[1] <- 0
  for (n in 2:length(t_vals)) {
```

```

    v <- t_vals[2] - t_vals[1]
    A_vals[n] <- A_vals[n - 1] + v * exp_model(A = A_vals[n -
        1], K = K, r = r, c = c)
}
soln <- r/c - (r/c - A_vals[1]) * exp(-c * t_vals)
t_val <- seq(h, 3, by = 0.01)
A_val <- rep(NA, times = length(t_val))
A_val[1] <- A_vals[length(A_vals)]
for (n in 2:length(t_val)) {
    v <- t_val[2] - t_val[1]
    A_val[n] <- A_val[n - 1] + v * exp2_model(A = A_val[n -
        1], c = c, K = K)
}
soln2 <- A_val[1] * exp(-c * (t_val - h))
x <- c(t_vals, t_val)
y <- c(A_vals, A_val)
plot(x, y, type = "l", col = "darkgreen", xlab = "Time (hours)",
    ylab = "Amount of Drug in Body (mg)", main = sprintf("Ibuprofen Model with r = %.1f",
        r), xlim = range(x), ylim = range(y))
abline(v = h, col = "red", lty = 2)
}

```



```
par(mfrow = c(1, 1))
```

Section 6: Appendix: Individual Contributions

Haoyang Huang(400302198): Part1_introduction

Chuhao Zhou(400384403): Part2_modeling

Bowen Zheng(400314765): Part3_result

Yucheng Wang(400386893): Part4_discussion

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